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Phase Difference and the Nature of Force — Derivation of the Range Formula in Void Phase Dynamics

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Phase Difference and the Nature of Force — Derivation of the Range Formula in Void Phase Dynamics

I. Mathematical Expression of Void Breaking

Void breaking is described as an impulse process, whose oscillation sequence satisfies:

$$[\sum \text{all impulses} = 0]$$

and the impulses must satisfy one of the following phase relations:

$$[\{+1, 0, -1, 0\}]$$

or its cyclic permutations, where $+1$ and -1 denote only the direction of energy borrowing (positive and negative energy), not the magnitude of energy.

The oscillation sequences in space can be expressed as $\{+1, 0\}$ and $\{-1, 0\}$, or $\{-1, 0\}$ and $\{+1, 0\}$. The oscillation sequence in time is $\{+1, 0, -1, 0\}$ or $\{-1, 0, +1, 0\}$, which forms the basis of three-valued resonance.

II. Two Modes of Spatial Convergence: In-phase Merging and Anti-phase Annihilation

Void breaking gives rise to space, and space possesses an inherent tendency to converge back to the void. This convergence proceeds through two pathways:

In-phase Merging: When two energy clusters of the same sign (both positive or both negative) meet, their violation of void symmetry points in the same direction. Space drives them to merge through topological contraction, reducing the total surface area, narrowing the scope of difference, and gently approaching the void.

Anti-phase Annihilation: When two energy clusters of opposite sign (one positive, one negative) meet, their violation of void symmetry points in completely opposite directions. Space has no room for compromise; it can only let them cancel each other out directly, zeroing out and fully restoring the void.

III. Two Correction Modes Driven by Phase Difference

When two oscillations (wave packets/particles) exist on the void substrate, their respective phases are θ_A and θ_B , and the phase difference is defined as:

$$[\Delta\theta = \theta_B - \theta_A]$$

The void does not permit unsettled deficits in the energy ledger, nor does it allow spatial surface energy to exist in non-minimal configurations. Therefore, the void automatically initiates a phase correction mechanism to eliminate the phase difference and synchronise the two oscillations. This correction process is what we perceive as "force."

Mode 1: Attraction (when $\Delta\theta < \pi$)

When the phase difference is less than π , the void judges that these two oscillations "should be pulled back into synchrony." The correction strategy is in-phase merging: bring them closer; the smaller the phase difference, the more balanced the energy and the smaller the spatial surface area. This process manifests macroscopically as **attraction**.

Mode 2: Repulsion (when $\Delta\theta > \pi$)

When the phase difference exceeds π , the void judges that "convergence through merging is no longer feasible" — the energy barrier to pull them back into synchrony exceeds the cost of letting them separate completely into the next synchronisation cycle. The correction strategy switches to "drive toward anti-phase annihilation": push them apart until the phase difference reaches π , at which point positive and negative energies directly cancel. This process manifests macroscopically as **repulsion**.

Extreme Case: $\Delta\theta = \pi$ (Perfect Anti-phase)

When the phase difference is exactly π , the positive half-cycle of one oscillation aligns precisely with the negative half-cycle of the other; they cancel directly, the energy ledger is instantly zeroed, and the local region of space completely reverts to the void. This corresponds to an **annihilation event**.

IV. Derivation of the Range Formula from Phase Difference

Let the masses of two particles (wave packets) be m_1 and m_2 , with corresponding intrinsic frequencies ν_1 and ν_2 . The number of phase conflicts borne by the void substrate between them accumulates with distance.

During the propagation time $t = r/c$, the numbers of rebirth cycles completed by the two particles are:

$$[N_1 = \nu_1 \cdot \frac{r}{c}, \quad N_2 = \nu_2 \cdot \frac{r}{c}]$$

The cumulative phase conflict between them is:

$$[\Delta N = |N_1 - N_2| = \frac{r}{c} \cdot |\nu_1 - \nu_2|]$$

Each unit conflict contributes a unit stress σ . When the accumulated stress reaches the void's rupture limit Σ_{crit} , the void ruptures (i.e., an interaction occurs):

$$[\sigma \cdot \frac{r_{\text{crit}}}{c} \cdot |\nu_1 - \nu_2| = \Sigma_{\text{crit}}]$$

Solving for the critical distance (the interaction range):

$$[r_{\text{crit}} = \frac{\Sigma_{\text{crit}}}{\sigma} \cdot \frac{c}{|\nu_1 - \nu_2|}]$$

Define the dimensionless constant:

$$[C \equiv \frac{\Sigma_{\text{crit}}}{\sigma}]$$

Then the range formula is:

$$[r_{\text{crit}} = C \cdot \frac{c}{|\nu_1 - \nu_2|}]$$

The physical meaning of C is: **the maximum number of rebirth conflicts the void can withstand before rupturing**. Experimental fitting gives $C \approx 0.1$, indicating that the void is extremely sensitive — less than a single complete beat conflict can trigger rupture.

V. Limiting Case of Frequency Matching ($\Delta\nu = 0$)

When the rebirth frequencies of two particles match exactly:

$$[\nu_1 = \nu_2 \quad \Rightarrow \quad \Delta\nu = 0]$$

Substituting into the range formula:

$$[\lim_{\Delta\nu \rightarrow 0} r_{\text{crit}} = \infty]$$

This means: on the void substrate, there exists no distance that would cause the void to rupture. The void tension between the two particles is perpetually zero; they can maintain a connection indefinitely without any external force — this is precisely the mechanical essence of **quantum entanglement and bound states**.

VI. Force Transmission Mechanism: Relay of Phase Settlement Signals

Force transmission does not occur through the exchange of some mysterious particle, but rather through **relay propagation of phase settlement signals and space-convergence pressure on the spacetime lattice**:

1. A change in particle A 's phase creates a local phase disturbance on the lattice, simultaneously producing an excess of local spatial surface energy — i.e., the space-convergence pressure has not yet been released.
2. This disturbance propagates at light speed c to neighbouring lattice points; each lattice, upon receiving the information, updates its own phase state and passes the convergence pressure downstream.
3. When the information reaches particle B , the latter adjusts its own oscillation according to the received phase information — if $\Delta\theta < \pi$, it is pulled closer (attraction); if $\Delta\theta > \pi$, it is pushed away (repulsion).
4. Photons, gluons, and other gauge bosons are not "force carriers"; they are energy-settlement by-products triggered by the phase-settlement signals and space-convergence pressure during different convergence operations — they are by-products of space executing its convergence, not the carriers of force.

VII. The Special Status of Gravity: The $\Delta\nu \rightarrow 0$ Limit

Gravity occupies a special position among all interactions:

- Its range is infinite ($r_{\text{crit}} \rightarrow \infty$), yet its strength is extremely weak.
- In the PDV framework, this arises because gravity corresponds to a frequency difference $\Delta\nu$ approaching zero.

Gravity is not a resonance phenomenon of some specific particle, but the **macroscopic accumulation of the phase gradient produced by the baseline beat shared by all particles within the World Phase**. The phase-correction contribution of each individual particle is negligible, but when a large number of particles aggregate into a celestial body, their collective effect manifests as gravity.

In other words:

$$[\text{Gravity} = \lim_{\Delta\nu \rightarrow 0} \left(C \cdot \frac{c}{\Delta\nu} \right) = \text{coherent amplification of the baseline beat of all existence}]$$

Gravity is the shadow of "existence" itself — as long as matter exists, space has not completely converged back to the void, and residual space continuously exerts a faint convergence pressure. This may be the deep reason why gravity has stubbornly resisted unification with the other three forces and quantization: it is not a specific force, but the global residue of the void's phase-correction and space-convergence mechanisms in the $\Delta\nu \rightarrow 0$ limit.

VIII. Supplementary: Phase-based Explanations of Cosmological Puzzles

Phenomenon	PDV Phase Explanation
Dark Matter	Matter that lies within the same phase cycle as our World Phase but outside the Electromagnetic Phase window. It participates in gravity (sharing the baseline beat) but is electromagnetically invisible.
Dark Energy	Antimatter (the negative half-cycle matter). The total positive and negative matter in the universe is strictly symmetric, with total energy zero. We do not see antimatter simply because our World Phase happens to reside in the positive half-cycle.
Cosmic Expansion	A meaningless concept. The universe is infinite; discussing the expansion of an infinite space is itself meaningless. The observed "recession" is a spurious effect arising from the phase reversal of gravity (switching from attraction to repulsion) when spatial scales exceed the gravitational wavelength.
Redshift	Two sources: cumulative phase deviations from successive energy settlements as photons traverse multiple World Phases; and energy loss due to the long-range phase reversal of gravity.
Time Dilation	High-speed motion means traversing more lattices per unit time; the annihilation-rebirth rhythm is perturbed, producing a "frame-dropping" effect. Cumulative phase deviation makes the apparent time slower.

Summary

The core logical chain of the above derivation is:

Void breaking → **impulse sequence** $\{+1, 0, -1, 0\}$ → **space convergence** (in-phase merging / anti-phase annihilation) → **phase-difference $\Delta\theta$ -driven correction** → **range formula** $r_{\text{crit}} = C \cdot c/|\Delta\nu|$ → **gravity as the $\Delta\nu \rightarrow 0$ limit** → **unified explanation of dark matter, dark energy, redshift, expansion, etc.**

All content introduces no extra-physical assumptions; only a single simple rule of the void substrate — **matching beats are stable, clashing beats break** — suffices to encompass all interactions from quantum entanglement to cosmic scales.

Translated by DeepSeek